



Indian Institute of Technology, Patna M.Tech. (Materials Science and Engineering)

Placement Brochure 2016-17



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About us

- Materials have played a key role in the development of mankind. Materials Science and Engineering (MSE) is an interdisciplinary field of science and engineering which investigates how changes in the structure of a material influence its properties. This field encompasses mechanical, chemical, biomedical, civil, electrical, and aerospace engineering physics; and chemistry.
- It is a discipline that enables both the creation and application of materials in society. Materials scientists and engineers develop materials for new applications, improve existing materials to enhance performance and evaluate ways in which different materials can be used together

Course Structure

**First Year
Specializations and Electives**



**Second Year
Research and Teaching
Assistantship**

Courses Offered

Core Courses

- Nano-structured Materials
- Materials Processing Technology
- Advanced Materials Characterization Techniques
- Structural and Functional Properties of Materials

Elective Courses

- Surface Engineering
- Rubber Science and Technology
- Advanced Building Materials
- Composite Science and Technology
- Alloy Development and Heat Treatment

Lab Courses

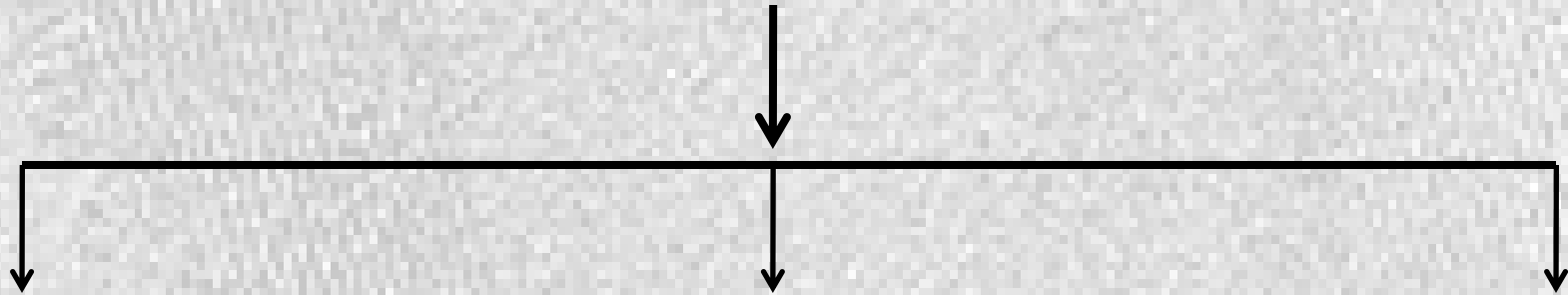
- Microstructure and Phase Analysis Laboratory
- Materials Characterization Laboratory

Research Work

Students undertake projects pertaining to some real time problems in final year under the guidance of distinguished faculties and complete their thesis as a part of the course.

LABORATORY FACILITIES

Ceramics and nano materials lab



Nano materials lab

Materials chemistry
lab

Ceramics lab



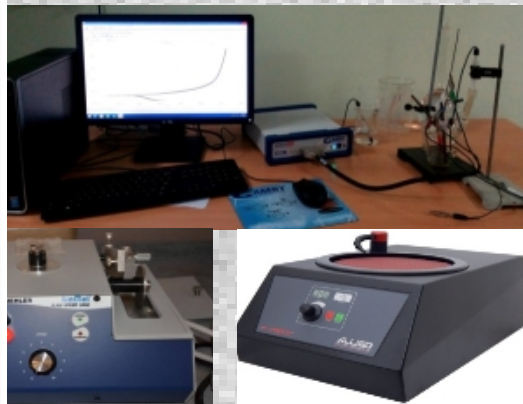
Ceramic and nano materials lab in department of MSE focuses on the area of materials chemistry of advanced materials which probes into the synthesis issues for the nano particles (via materials chemistry routes) and try to understand the complexities involved via different structural characterisation methods.

Metallurgical and materials engineering lab

Plasma spray lab

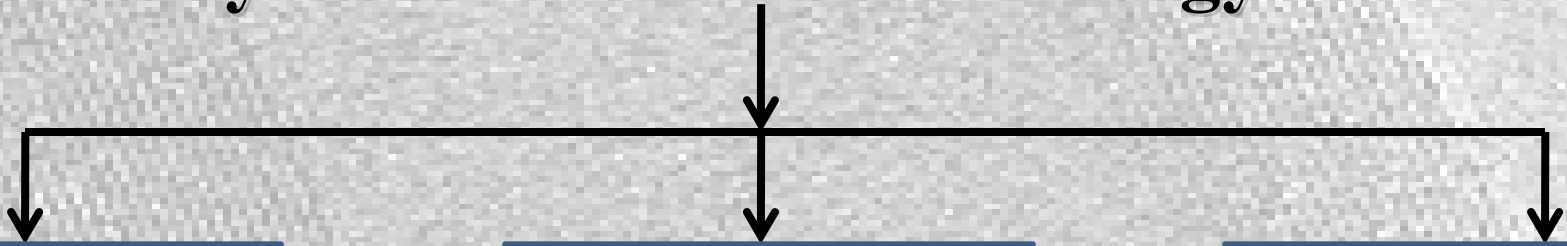
Metallographic and corrosion lab

Mechanical testing lab



Metallurgical and materials engineering lab in department of MSE has set-up the world-class plasma spray coating facility (Oerlikon Metco, USA), which has the potential of fabricating the large domain of metal and ceramic coating for the wear and corrosion resistance applications. In addition, sophisticated micro indentation with the scratch tester (Microtest, Spain) is a part of mechanical testing lab. For characterizing the corrosion behaviour of coating/substrate, corrosion lab consisting of Potentiostat/Galvanostat (GAMRY, USA) has been established

Polymer science and technology lab



**Polymer
characterization lab**



Polymer processing lab



Polymer synthesis lab



Polymer science and technology lab in department of MSE has set-up the world-class polymer synthesis, processing and testing lab.

Ongoing research activities:- Projects related to polymer science and technology with specialization in adhesion, blends / alloys, composites (micro and nano), hybrid fillers (conventional / nano), thermal stability and chemical modification of polymers

Student Profile



The admissions to M.Tech. in Materials Science and Engineering program is based on the performance of students in the national level GATE (Graduate Aptitude Test in Engineering) examination along with personal interview. The students are chosen from different engineering backgrounds viz. Mechanical Engineering, Plastic Technology, Chemical Engineering, Metallurgy.

The M.Tech. degree is designed in a way to provide a holistic view on all the classes of materials including metals, ceramics and polymers. The program is intended to provide in-depth knowledge in the fundamentals, analysis and structure-property correlation of various materials system. Current batch of Materials Science and Engineering is working on following projects

- Ceramic nano composites.
- Renewable processing oils for the compounds used in tyre.
- Polymer blends and composites.



Faculty Profile



Dr. Dinesh Kumar Kotnees

**Head: Department of Materials
Science & Engineering**

Email id : dinesh@iitp.ac.in

Research Areas : Polymer Science and Technology with specialization in Adhesion, Blends, Composites, Fillers and Bulk/Surface properties of Polymers



Dr. Anirban Chowdhury

Asst. Professor

Email id : anirc@iitp.ac.in

Research Area :

Materials Chemistry - chemical synthesis - structural and spectroscopic characterisations - thin films & coatings - nanomaterials- sol gel - ceramics



Dr. Anup Kumar Keshri
Asst. Professor

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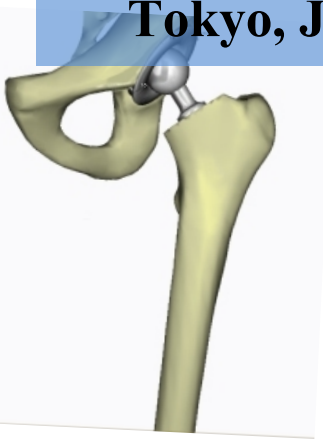
Research Area : Carbon Nanotube Reinforced Ceramic Matrix and Metal Matrix Composites, Thermal Spraying, Tribology of Materials, Process-Structure-Property Relationship

Sponsored Projects

Denka



- **Fabrication of Robust Plasma Sprayed Rare Earth Oxide Hydrophobic Coating for the High Temperature and Wear Resistance Applications.(Sponsored by SERB-DST)**
- **Synthesis & characterization of faceted nano crystalline powders of Ceria-Zirconia and related systems. .(Sponsored by SERB-DST)**
- **Surface modified metallic ortopedic implant for sustained drug release. .(Sponsored by DST/TSG/AMT)**
- **Simultaneous improvement of low temperature and room temperature properties of elastomers. .(Sponsored by DENKA, Tokyo, Japan)**



Collaborations



JK Tyres	NCL Pune
BHEL Hyderabad	IRMRA Mumbai
NML Jamshedpur	Action Shoes Delhi



Several students have also gone at these reputed organisations for Internships and M.Tech projects



Placements

IIT Bombay	IIT Patna	RRB
TATA Steel	DRDO	PWD
	IIT Kharagpur	



Several students also choose to go for higher studies. Many students from the M.Tech Materials Science and Engineering program have also been selected to visit various international laboratories/institutions/universities

Contact Us

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